For this Project Based Learning assignment, my team and I built a production-ready Digital Twin of a V6 Internal Combustion Engine in AutoCAD 3D. The focus was not just on shapes — it was on mechanical integration and engineering precision at every step.

- **Surface Modeling** — Used LOFT and SWEEP commands to model intake manifolds and complex piping — geometries that standard extrusion simply could not achieve.
- **Multi-Axis Alignment** — Mastered the User Coordinate System to position pistons along 60-degree cylinder banks, maintaining geometric symmetry through POLAR ARRAY operations.
- **Design for Manufacturing** — Executed manual coring to define accurate wall thicknesses for casting feasibility and applied Boolean Logic to carve out coolant passages and cylinder bores.
- **Assembly & Interference** — Managed 12+ unique components by separating moving parts from the static block via layers, then ran interference checks to ensure zero collisions between the crankshaft and connecting rods.

**Skills Applied:** AutoCAD 3D Modeling · Advanced Surfacing · Boolean Logic · Multi-Axis UCS · Interference Detection · Design for Manufacturing